

Wireless Access Electromagnetics

Homework 3

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Problem 1:

Problem 1

$$f(\theta, \varphi) = \begin{cases} 1 & \text{for } \theta < \theta_0 \\ 0 & \text{for } \theta > \theta_0 \end{cases}$$

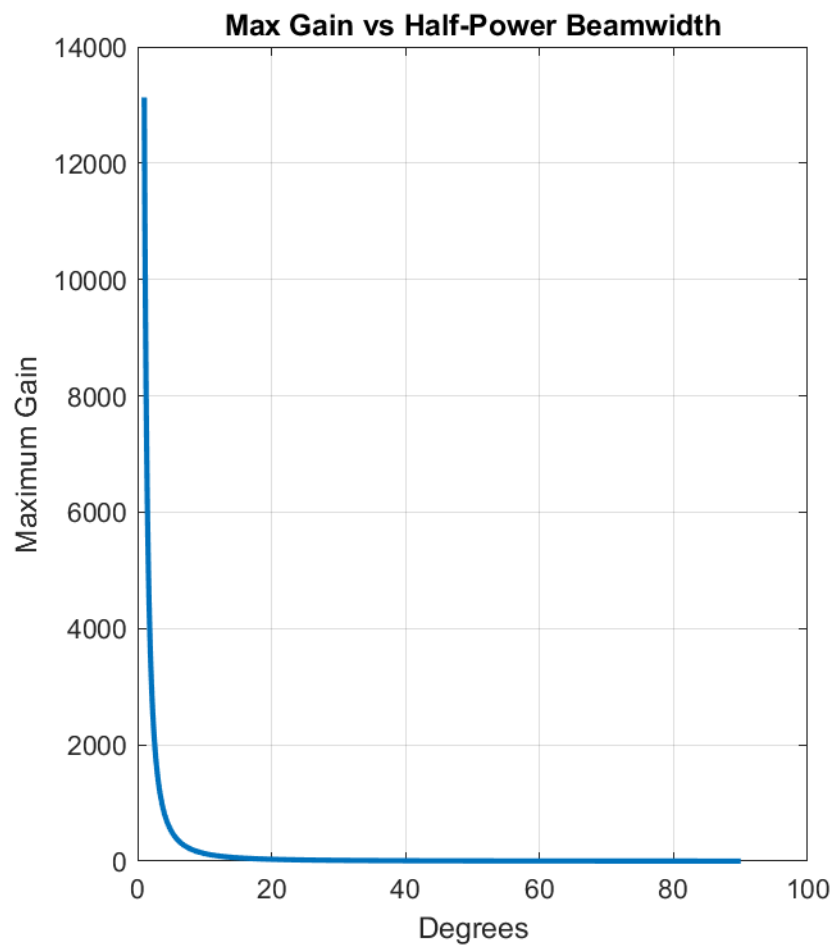
Sol: $D = \frac{4\pi}{\Omega_A}$, $\Omega_A = \int_0^{2\pi} \int_0^\pi f(\theta, \varphi) \sin\theta d\theta d\varphi$

$$\Omega_A = \int_0^{2\pi} d\varphi \int_0^{\theta_0} \sin\theta d\theta = \left[2\pi [-\cos\theta]_0^{\theta_0} \right] = 2\pi (1 - \cos\theta_0)$$

$$\Omega_A = 2\pi (1 - \cos\theta_0)$$

$$G(\theta_0) = \frac{4\pi}{2\pi (1 - \cos\theta_0)} \Rightarrow \frac{2}{(1 - \cos\theta_0)}$$

This formula demonstrates the fundamental antenna trade-off: higher Gain is achieved by narrowing the Beamwidth to focus most of the radiated power into a smaller solid angle.



This results shows the inverse relation between the Gain and the beamwidth, where increasing the Beamwidth decreases the Gain exponentially and vice-versa. The Gain G reaches a peak of approx: 13300 at 1 degree and decreasing to almost 0 at 90 degrees.

Problem 2:

Problem 2:

Sol: $L/\lambda_0 = 0.23$, $W/\lambda_0 = 0.52$

$$E_{\text{norm}}(\theta, \varphi) = \left| \sin \theta \sin \left(\pi \frac{W}{\lambda_0} \cos \theta \right) \cdot \cos \left(\pi \frac{L}{\lambda_0} \sin \theta \sin \varphi \right) \right|$$

For x - y plane, $x \geq 0$:

$$\varphi \in [-\pi/2, \pi/2], \theta = \frac{\pi}{2}$$

$$E_{\text{norm}} = \left(\frac{\pi}{2}, \varphi \right) = \left| \cos \left(\pi \frac{L}{\lambda_0} \sin \varphi \right) \right|$$

For x - z planes $x \geq 0$

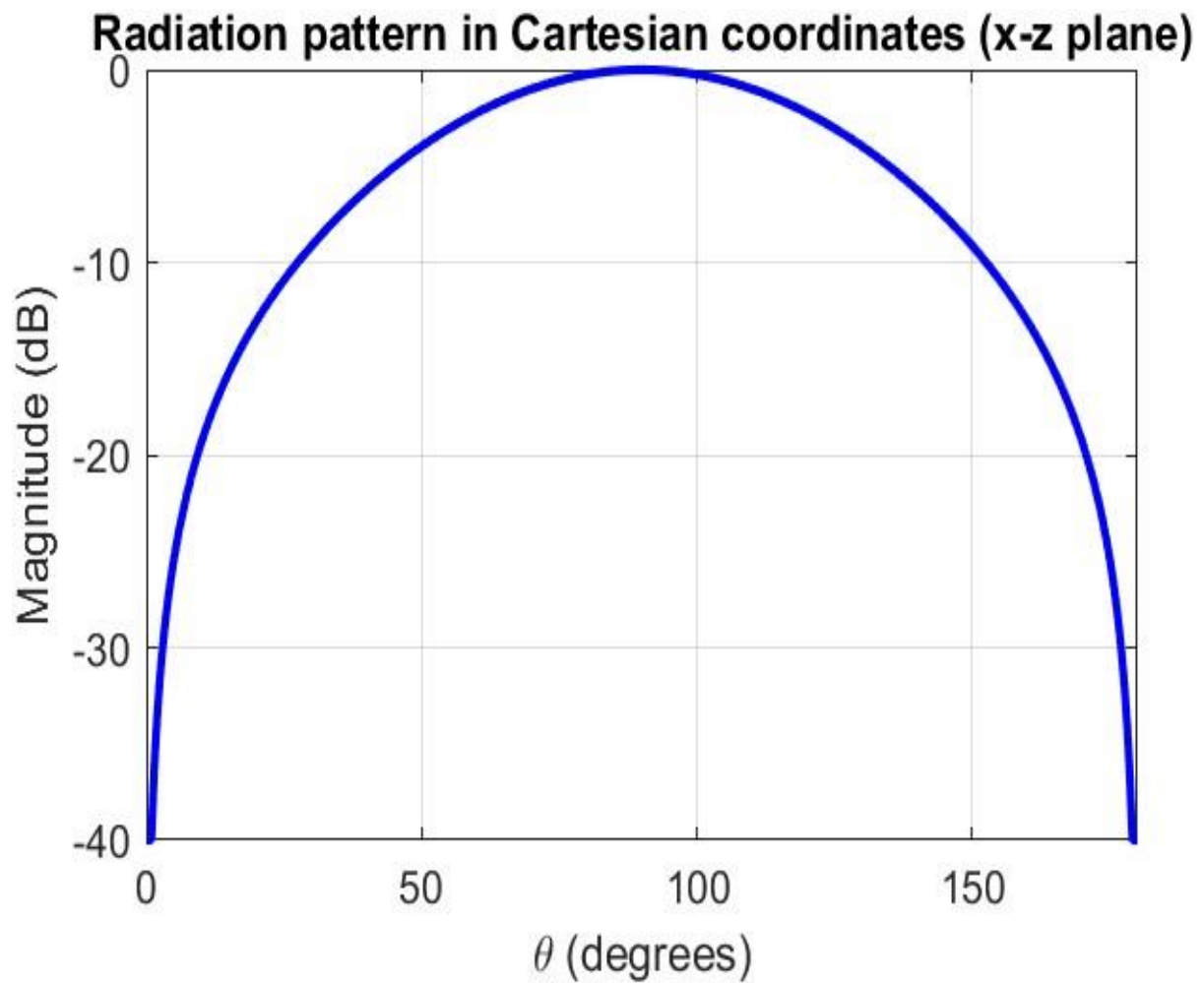
$$\varphi = 0$$

$$\theta = [0, \pi]$$

$$E_{\text{norm}}(\theta, 0) = \left| \sin \theta \sin \left(\pi \frac{W}{\lambda_0} \cos \theta \right) \right|$$

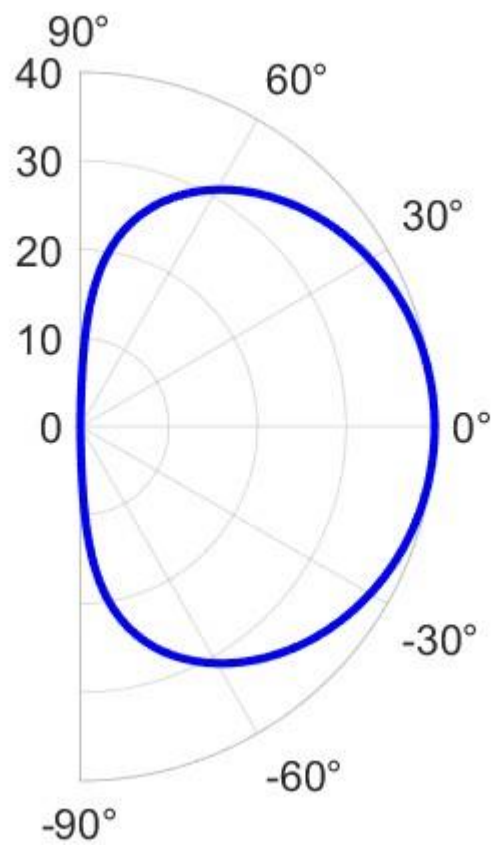
For x-z plane , $X \geq 0$

1- Cartesian



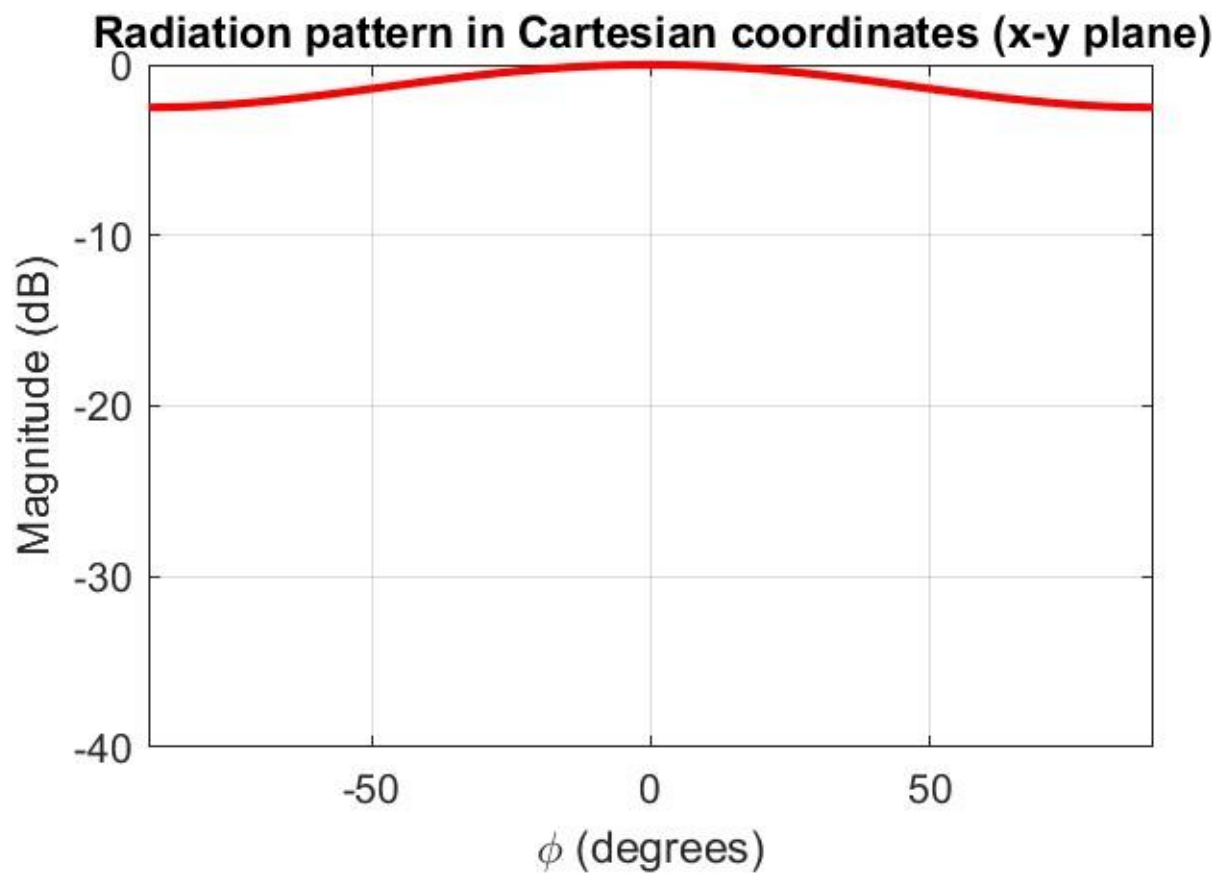
2- Polar format :

Radiation pattern in Polar coordinates (x-z plane)



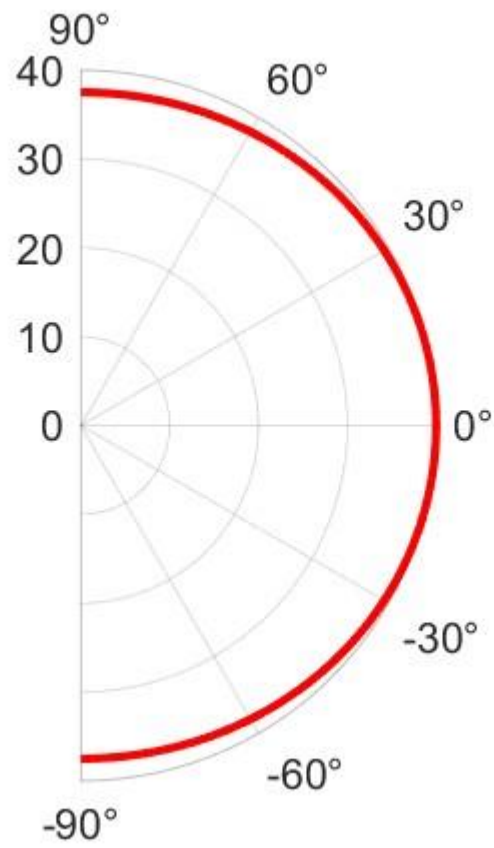
For x-y plane , $X \geq 0$

1- Cartesian plane



2- Polar form:

Radiation pattern in Polar coordinates (x-y plane)



Problem 3:

Prob 3:

Sol, $P_{rad} = \frac{1}{2} R_{rad} |I_0|^2$

$$P_{rad} = \iint_{\text{sphere}} S(r, \theta, \varphi) d\Omega = \frac{1}{2} \frac{1}{Z_0} \frac{1}{(4\pi r^2)} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin\theta |E(\theta, \varphi)|^2$$

$$R_{rad} = \frac{1}{|I_0|^2} \frac{1}{Z_0} \frac{1}{(4\pi)^2} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin\theta |E(\theta, \varphi)|^2$$

$$|E(\theta, \varphi)|^2 = 4Z_0^2 |I_0|^2 (\sin\theta)^{3/2}$$

$$\Rightarrow \frac{1}{|I_0|^2} \frac{1}{Z_0} \frac{1}{(4\pi)^2} \cdot 2\pi \int_0^\pi \sin\theta \cdot 4Z_0^2 |I_0|^2 (\sin\theta)^{3/2} d\theta$$

$$\Rightarrow \frac{1}{|I_0|^2} \frac{1}{Z_0} \frac{1}{(4\pi)^2} \cdot 2\pi \int_0^\pi 4 |I_0|^2 Z_0^2 \sin^4\theta d\theta$$

$$\Rightarrow \frac{1}{16\pi^2} 8\pi Z_0 \int_0^\pi \sin^4\theta d\theta = \frac{1}{2} 420\pi \left(\frac{3}{8} \theta - \frac{\sin 2\theta}{4} + \frac{\sin 4\theta}{32} \right) \Big|_0^\pi$$

$$R_{rad} = 60 \left(\frac{3}{8} \pi \right) = 70.68 \Omega$$

$$2) P_{rad} = \frac{1}{2} R_{rad} |I_0|^2 = \frac{1}{2} 70.68 |I_0|^2$$

$$P_{rad} = 35.34 |I_0|^2$$

Problem 4:

Ex 4:

$$e(\theta, \varphi) = \frac{\left(\frac{dP}{d\Omega} \right)(\theta, \varphi)}{\frac{P_{in}}{4\pi r^2}} = \frac{S(r, \theta, \varphi)}{\frac{P_{in}}{4\pi r^2}} \therefore S(r, \theta, \varphi) = \left(\frac{dP}{d\Omega} \right)(\theta, \varphi)$$

$$\Rightarrow \frac{1}{2} \frac{1}{Z_0} \frac{1}{(4\pi r^2)} |e(\theta, \varphi)|^2 = \frac{P_{in}}{4\pi r^2}$$

$$|e(\theta, \varphi)| = \sqrt{2 Z_0 4\pi} e(\theta, \varphi) P_{in}$$

$$|e(\theta_{max}, \varphi_{max})| = \sqrt{2 Z_0 4\pi G P_{in}} = \sqrt{2 \cdot 120\pi \cdot 4\pi \cdot 6309.57 \cdot 12}$$

$$= \frac{26,789}{10,000} \approx 0.212 \frac{V}{m}$$

Problem 5:

Ex 5,

Sol, $P_{av}^{max} = \left(\frac{dP}{d\xi} \right) \cdot a_{eq}$

$$a_{eq} = G \cdot \frac{1^2}{4\pi}, \quad \left(\frac{dP}{d\xi} \right) = \frac{1}{2} \frac{|E_{inc}|^2}{Z_0}$$

$$P_{av}^{max} = G \cdot \frac{1^2}{4\pi} \times \frac{1}{2} \frac{|E_{inc}|^2}{Z_0} = \frac{G \cdot 1^2 \cdot |E_{inc}|^2}{8\pi Z_0}$$

$$P_{av} = \frac{|V_a|^2}{8 \operatorname{Re}\{Z_{Rx}\}} = \frac{G \cdot 1^2 \cdot |E_{inc}|^2}{8\pi Z_0}$$

$$|V_a|^2 = \frac{G \cdot 1^2 \cdot |E_{inc}|^2 \cdot \operatorname{Re}\{Z_{Rx}\}}{\pi Z_0}$$

$$|V_a|^2 = \frac{316.2 \times (0.03)^2 \times (0.002)^2 \times 75}{\pi \times 377}$$

$$|V_a|^2 = 7.21 \times 10^{-8} \text{ V}^2$$

$$|V_a| = 2.69 \times 10^{-4} \text{ V}$$